1. Design and implement a segmentation algorithm for Regions of Interest (ROI) in medical images (either organs or lesions) based on an analytical pipeline including a preprocessing step (e.g. image denoising, image filters to enhance the shape of the lesions) and a series of rules coded to identify the ROI mask. Evaluate the robustness of your segmentation pipeline in terms of standard segmentation quality metrics (e.g. Dice, Jaccard indices) for different settings in the space of free parameters of your algorithm.
   Suitable datasets:

- 2D data (Mammography, RX) and 3D data (CT, MRI) can be found in The Cancer Imaging Archive (TCIA) repository: https://www.cancerimagingarchive.net/access-data/
- A curated dataset of mammograms with annotations is described in https://www.nature.com/articles/sdata2017177 and made available at https://wiki.cancerimagingarchive.net/display/Public/CBIS-DDSM
- A small dataset of mammograms with annotations is available within the course materials*****: DATASET/IMAGES/Mammography_masses/